Momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate and compare signed momentum.

SUCCESS CRITERIA - I can...

1. Multiply mass by velocity with correct units.
2. Include direction or a sign.
3. Distinguish zero total momentum from no motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An asteroid's crossing window is days **13**-**17**; Earth's is **15**-**19**. **Find the overlap** and **explain why this** is not proof of a collision.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $p = mv$; $p_{total} = p_1 + p_2$.
Use signed velocities; momentum units are kg·m/s.

Worked reasoning: A **3 kg** cart moving left at **2 m/s** has momentum **-6 kg·m/s** when right is positive. The minus sign gives direction, not a negative mass.

Your turn: Draw separate velocity and momentum arrows, labeling their different units.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The two carts have the same speed in opposite directions. Which momentum values are positive if right is positive?



2 kg; 3 m/s right

2 kg; 3 m/s left

Marzano check - circle: ① ② ③

04 Apply the physics

A **0.50 kg** cart moves at **+6 m/s**; a **1.5 kg** cart moves at **-2 m/s**.
1. **Find each momentum** and total momentum.
2. **Draw the momentum arrows** and **explain why zero total** does not mean both carts are at rest.

Object / trial	Mass	Velocity	Momentum
A	___	___	___
B	___	___	___

Marzano check - circle: ① ② ③

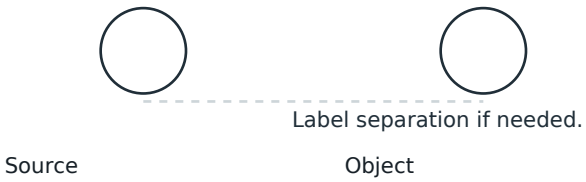
05 Explain your reasoning

One cart doubles mass at the same velocity. Another doubles speed at the same mass. **Compare the momentum multipliers**.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Outside a spherical source, field at r is **144 N/kg**. **Find field** at **6** r and the force on **0.8 kg** there.
Label the masses, field or force relevant to the question.



Marzano check - circle: ① ② ③

07 Exit check

A cart has momentum **-12 kg·m/s** and mass **3 kg**. **Find velocity**. Then **calculate its momentum** if velocity reverses without changing speed.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Three carts have masses **1**, **2**, **3 kg** and velocities **+4**,**-5**, v m/s. **Choose v** for zero total momentum. Then reverse only the **1 kg** cart and recalculate the total.

Object / trial	Mass	Velocity	Momentum
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③